

TURKU AP, Finland

WMO: 029720

Lat: 60.516N

Lon: 22.279E

Elev: 59

StdP: 100.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.0	-17.8	-23.2	0.5	-20.7	-19.9	0.6	-17.4	9.7	-2.2	8.7	-1.5	2.0	70	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	27.1	18.0	25.1	17.3	23.2	16.3	19.7	24.4	18.6	22.9	17.5	21.7	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.3	17.0	12.2	20.3	15.9	11.3	19.4	56.3	24.4	52.6	22.8	49.4	21.7	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.7	-23.8	29.1	3.7	2.0	-26.5	30.6	-28.6	31.7	-30.7	32.8	-33.5	34.3		
(5)			-23.8	20.8	3.8	1.3	-26.5	21.7	-28.7	22.5	-30.8	23.2	-33.5	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-4.5	-4.9	-1.6	4.2	9.6	14.0	17.3	16.0	11.3	5.7	1.4	-1.8
	DBStd	8.77	5.87	6.05	4.31	3.43	3.95	3.24	2.97	2.97	3.30	3.96	4.44	5.50
	HDD10.0	2250	450	418	359	177	54	3	0	1	24	139	258	367
	HDD18.3	4682	708	651	617	424	271	135	56	84	210	391	508	625
	CDD10.0	652	0	0	0	3	43	123	226	186	64	7	0	0
	CDD18.3	42	0	0	0	0	1	5	23	12	0	0	0	0
	CDH23.3	399	0	0	0	0	21	62	223	93	1	0	0	0
	CDH26.7	70	0	0	0	0	2	7	49	12	0	0	0	0
Wind	WSAvg	3.2	3.3	3.2	3.2	3.2	3.2	3.3	3.0	2.9	3.0	3.2	3.4	3.4
Precipitation	PrecAvg	723	60	44	42	39	36	50	73	82	69	72	79	70
	PrecMax	964	108	106	103	102	89	105	146	190	163	173	163	182
	PrecMin	531	9	3	4	2	4	7	0	0	12	21	12	7
	PrecStd	96	27	25	22	22	23	27	41	44	36	32	38	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	5.1	10.9	18.9	26.0	27.8	30.2	28.2	22.0	15.1	10.1	8.1
		MCWB	4.8	3.3	5.7	11.1	16.4	17.4	19.5	18.3	15.6	12.3	9.5	7.4
	2%	DB	4.1	3.9	7.8	15.8	22.9	25.2	28.0	25.9	19.9	13.2	9.0	6.9
		MCWB	3.6	2.9	4.3	9.1	14.1	16.2	18.6	17.9	15.1	11.6	8.4	6.5
	5%	DB	3.1	2.9	5.2	13.2	20.2	23.0	25.9	24.0	18.2	12.2	8.1	5.8
		MCWB	2.6	2.3	2.8	7.6	12.9	15.3	17.9	17.2	14.2	10.8	7.5	5.2
	10%	DB	2.1	2.0	3.8	11.0	17.4	20.9	23.9	22.1	16.9	11.2	7.1	4.8
		MCWB	1.7	1.5	1.9	6.4	11.3	14.2	17.2	16.3	13.8	10.0	6.6	4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	3.8	6.4	11.6	17.7	19.5	21.3	20.7	16.9	13.3	9.6	7.5
		MCDB	5.2	4.3	9.3	17.1	22.7	25.1	26.6	24.9	19.8	14.2	9.8	7.7
	2%	WB	3.6	2.9	4.4	9.9	15.6	17.5	20.2	19.4	15.9	12.2	8.5	6.3
		MCDB	3.9	3.3	6.6	14.7	21.0	22.8	25.6	23.3	18.7	13.2	8.8	6.6
	5%	WB	2.6	2.1	3.2	8.3	13.7	16.1	19.0	18.3	15.1	11.3	7.5	5.1
		MCDB	2.9	2.6	4.7	12.2	19.1	20.9	23.6	22.0	17.4	12.2	7.9	5.4
	10%	WB	1.6	1.3	2.2	6.8	11.8	15.0	17.9	17.3	14.2	10.4	6.6	3.9
		MCDB	1.9	1.7	3.3	10.2	16.9	19.5	22.4	20.9	16.3	11.3	7.0	4.3
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.9	7.4	9.2	10.3	9.5	9.5	8.4	6.1	4.4	4.8	
		MCDBR	4.4	5.0	8.8	13.5	14.5	13.4	12.9	12.2	9.9	6.3	4.3	4.4
		MCWBR	4.3	4.4	6.1	7.9	7.4	6.4	5.5	5.5	5.8	4.8	4.2	4.3
	5% WB	MCDBR	4.1	4.5	7.5	12.0	12.6	11.2	10.9	9.6	8.4	5.1	4.2	4.1
		MCWBR	4.1	4.2	5.6	7.5	7.0	6.0	5.4	5.1	5.4	4.3	4.3	4.1
Clear-Sky Solar Irradiance	taub	0.273	0.311	0.331	0.370	0.366	0.358	0.386	0.376	0.347	0.330	0.298	0.382	
	taud	2.148	2.236	2.316	2.327	2.403	2.450	2.394	2.417	2.485	2.400	2.210	2.614	
	Ebn at Noon	471	671	792	825	860	873	834	813	777	653	434	325	
	Edh at Noon	41	68	87	104	104	101	105	95	75	59	39	28	
All-Sky Solar Radiation	RadAvg	0.24	0.82	2.33	3.89	5.29	5.61	5.26	4.13	2.53	1.10	0.30	0.14	
	RadStd	0.04	0.16	0.28	0.38	0.50	0.44	0.52	0.42	0.31	0.18	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A	N/A

Nomenclature: See separate page